CS315 · ALGORITHMS ANALYSIS & DESIGN

# Practice 4 الحل

DP و Lecture 4 · Greedy و 5

**محارب**

## 01 حل Q1 و Q2

| # | الجواب | السبب |
|---|---|---|
| 1 | **True** | |
| 2 | **False** | `O(2ⁿ)` . الـ DP هو اللي `O(n)` . |
| 3 | **False** | الـ Greedy ما يرجع عن قراره. |
| 4 | **False** | مثال السلايد: 160 بدل 220. |
| 5 | **True** | |
| 6 | **True** | |

Q2: 1. c (10·20·5) 2. a 3. b 4. b

## Q3: Matrix chain حل 02

```
p = 30, 35, 15, 5, 10

AB = 30·35·15 = 15,750      BC = 35·15·5 = 2,625      CD = 15·5·10 = 750

A(BC) = 2,625 + 30·35·5 = 7,875    ✓

(AB)C = 15,750 + 30·15·5 = 18,000

B(CD) = 750 + 35·15·10 = 6,000

(BC)D = 2,625 + 35·5·10 = 4,375    ✓

ABCD: A(BCD)    = 4,375 + 30·35·10 = 14,875
      (AB)(CD) = 15,750 + 750 + 30·15·10 = 21,000
      (ABC)D    = 7,875 + 30·5·10 = 9,375    ✓
```

| | A | B | C | D |
|---|---|---|---|---|
| **A** | 0 | 15750 | 7875 | **9375** |
| **B** | — | 0 | 2625 | 4375 |
| **C** | — | — | 0 | 750 |
| **D** | — | — | — | 0 |

```
Minimum = 9,375        Order: ((A(BC))D)
```

## 03 حل Q4: LCS

| | ∅ | B | D | C | B |
|---|---|---|---|---|---|
| ∅ | 0 | 0 | 0 | 0 | 0 |
| B | 0 | **1** | 1 | 1 | 1 |
| A | 0 | 1 | 1 | 1 | 1 |
| C | 0 | 1 | 1 | **2** | 2 |
| D | 0 | 1 | 2 | 2 | 2 |
| B | 0 | 1 | 2 | 2 | **3** |

Length = 3  LCS = BCB  (BDB is also correct)

## 04 حل Q5 و Q6

### Q5

sorted by f already

take a1 (f = 3)

a2 (s = 2 < 3) ✗  a3 (s = 4 ≥ 3) ✓ take, f = 7

a4 (s = 1) ✗  a5 (s = 5 < 7) ✗

a6 (s = 8 ≥ 7) ✓ take, f = 10

a7 (s = 9 < 10) ✗

Answer: {a1, a3, a6} → 3 activities

## Q6

| الغرض | w | v | $\frac{v}{w}$ |
|---|---|---|---|
| **2** | 4 | 40 | 10 |
| **1** | 5 | 30 | 6 |
| **3** | 6 | 30 | 5 |
| **4** | 3 | 6 | 2 |

a) Fractional: item 2 (4 kg, 40) + item 1 (5 kg, 30)

+ $\frac{1}{6}$ of item 3 (1 kg, 5)

total = 40 + 30 + 5 = 75

b) 0-1 greedy: item 2 + item 1 = 70 (1 kg left; nothing else fits)

Best 0-1: item 2 + item 3 (10 kg) → 70

item 1 + item 2 (9 kg) → 70

So greedy gives 70 here, which happens to be optimal.

## Q7: Huffman حل 05

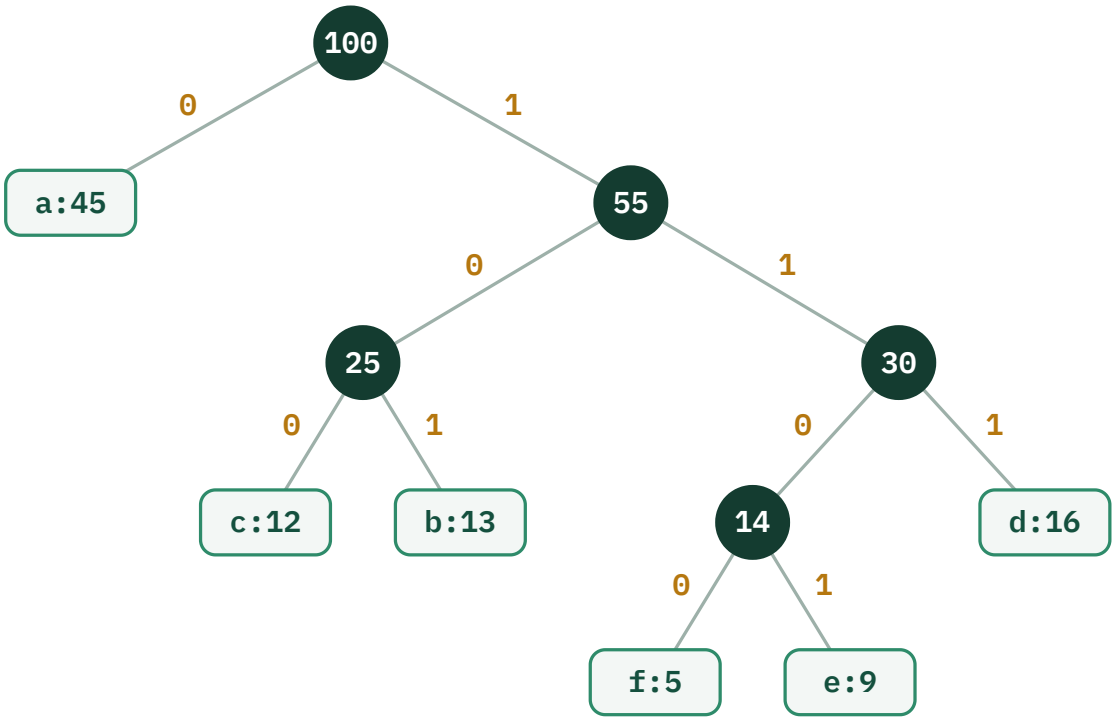
f:5 + e:9 = 14

c:12 + b:13 = 25

14 + d:16 = 30

25 + 30 = 55

a:45 + 55 = 100 (root)




| الحرف | التكرار | الكود | البتات |
|---|---|---|---|
| a | 45 | 0 | 45 |
| b | 13 | 101 | 39 |
| c | 12 | 100 | 36 |
| d | 16 | 111 | 48 |
| e | 9 | 1101 | 36 |
| f | 5 | 1100 | 20 |

```
Huffman: 45 + 39 + 36 + 48 + 36 + 20 = 224 bits
```

```
Fixed 3-bit code: 100 × 3 = 300 bits
```